

# IMDB Movie Analysis

## 1. PROJECT DESCRIPTION :

This project involves a comprehensive analysis of an IMDB movie dataset, covering various aspects such as movie genres, duration, language, directors, and budget. Tasks include determining the most common genres and their impact on IMDB scores, analyzing the relationship between movie duration and ratings, examining language distribution and its impact on scores, identifying top directors and their contribution, and exploring the correlation between budgets and gross earnings to identify movies with the highest profit margin. The analysis aims to unveil patterns and insights into the factors influencing movie success on IMDB.

## 2. APPROACH:

For this project, first, we'll get an understanding of the given data. Then We'll clean the data as per our requirement by removing null values, deleting unnecessary columns, etc. After the cleaning, we'll use various functions, and charts for desired answers to the questions.

**Cleaning the data:** This is one of the most important steps to perform before moving forward with the analysis. Use your knowledge learned till now to do this. (Dropping columns, removing null values, etc.)

Dropping unnecessary columns.

(Color, director\_facebook\_likes, actor\_3\_facebook\_likes, actor\_2\_name, actor\_1\_facebook\_likes, cast\_total\_facebook\_likes, actor\_3\_name, facenumber\_in\_posts, plot\_keywords, movie\_imdb\_link, content\_rating, actor\_2\_facebook\_likes, aspect\_ratio, movie\_facebook\_likes)

Remove Blank Cell / Null Value.

Removing Duplicate.

A. MOVIE GENRE ANALYSIS:

- The task was to Analyze the distribution of movie genres and their impact on the IMDB score:
  - i. Firstly the genres were split into separate columns using delimiters as '|’.
  - ii. Then used COUNTIF to calculate number of occurrences of each genre.

The formula =COUNTIF(P3:V3885, "\*Action\*") counts the number of cells within the range P3 to V3885 that contain the text "Action." It utilizes the COUNTIF function in Excel, where the first argument specifies the range of cells to be evaluated (P3:V3885), and the second argument sets the criteria for counting, here denoted by the asterisks (\*) as wildcard characters to include any cell containing the term "Action," regardless of its position within the cell. The formula provides a count of occurrences of the specified genre ("Action") in the given range.

- iii. After using the above formula to calculate for count of each genre.
  - iv. Then, for each genre, I have calculated descriptive statistics (mean, median, mode, range, variance, standard deviation) of the IMDB scores using the following formulas.
  - v. The most common genre of movies in the dataset is DRAMA with count 1958.

| Genre       | Count of Genre |
|-------------|----------------|
| Action      | 970            |
| Adventure   | 795            |
| Animation   | 199            |
| Biography   | 244            |
| Comedy      | 1511           |
| Crime       | 719            |
| Documentary | 64             |
| Drama       | 1958           |
| Family      | 452            |
| Fantasy     | 517            |
| Horror      | 397            |
| Musical     | 103            |
| Mystery     | 389            |
| Romance     | 885            |
| Sci-Fi      | 501            |
| Thriller    | 1127           |
| Western     | 60             |

## A. MEAN

The formula **=AVERAGEIF(P3:V3885,X3,N2:N5044)** calculates the average IMDb score for movies in a specified genre (given in cell X3) within the range P3:V3885. It considers the corresponding IMDb scores in the range N2:N5044 based on the matching genre criteria.

## B. MEDIAN

**=IF(SUMPRODUCT((P3:V3=X3)\*(ISNUMBER(N3))), N3, "")**

This formula checks if the genre in columns P to V matches X3 and if the corresponding IMDb score in column N is a number. If both conditions are true, it returns the IMDb score; otherwise, it returns an empty string.

**=MEDIAN(IF(W3:W3885<>"", W3:W3885))** here the column W indicates the helper column calculated above

This formula checks if the cells in column W are not empty (indicating a specific genre) and calculates the median for those IMDb scores.

## C. MODE

**=MODE(IF(W3:W3885<>"", W3:W3885))**

This formula checks if the cells in column W are not empty and calculates the mode for those values.

## D. RANGE

**=MAX(IF((P3:P3885=X3)+(Q3:Q3885=X3)+(R3:R3885=X3)+(S3:S3885=X3)+(T3:T3885=X3)+(U3:U3885=X3)+(V3:V3885=X3), N2:N5044)) - MIN(IF((P3:P3885=X3)+(Q3:Q3885=X3)+(R3:R3885=X3)+(S3:S3885=X3)+(T3:T3885=X3)+(U3:U3885=X3)+(V3:V3885=X3), N2:N5044))**

This formula calculates the range of IMDb scores for movies matching the specified genre in cell X3. It finds the maximum

IMDb score and subtracts the minimum IMDb score for that genre.

E. STANDARD DEVIATION

=STDEV(IF((P3:P3885=X3)+(Q3:Q3885=X3)+(R3:R3885=X3)+(S3:S3885=X3)+(T3:T3885=X3)+(U3:U3885=X3)+(V3:V3885=X3), N2:N5044))

This formula calculates the standard deviation of IMDb scores for movies matching the specified genre in cell X3.

F. VARIANCE

=VAR(IF((P3:P3885=X3)+(Q3:Q3885=X3)+(R3:R3885=X3)+(S3:S3885=X3)+(T3:T3885=X3)+(U3:U3885=X3)+(V3:V3885=X3), N2:N5044))

This formula calculates the variance of IMDb scores for movies matching the specified genre in cell X3.

| Genre       | Count of Genre | Mean        | Median | Mode | Range | Standard dev | Variance    |
|-------------|----------------|-------------|--------|------|-------|--------------|-------------|
| Action      | 970            | 6.461649485 | 6.5    | 6.7  | 7.9   | 1.125115866  | 1.265885711 |
| Adventure   | 795            | 6.467460317 | 6.5    | 6.7  | 7.9   | 1.125115866  | 1.265885711 |
| Animation   | 199            | 6.573913043 | 0      | 0    | 0     | -            | -           |
| Biography   | 244            | 6.461057692 | 0      | 0    | 0     | -            | -           |
| Comedy      | 1511           | 6.412644788 | 0      | 0    | 0     | -            | -           |
| Crime       | 719            | 6.495366795 | 0      | 0    | 0     | -            | -           |
| Documentary | 64             | 6.7625      | 0      | 0    | 0     | -            | -           |
| Drama       | 1958           | 6.399285714 | 0      | 0    | 0     | -            | -           |
| Family      | 452            | 6.566666667 | 6.5    | 6.7  | 7.9   | 1.125115866  | 1.265885711 |
| Fantasy     | 517            | 6.375675676 | 0      | 0    | 0     | -            | -           |
| Horror      | 397            | 6.457575758 | 0      | 0    | 0     | -            | -           |
| Musical     | 103            | 7.25        | 0      | 0    | 0     | -            | -           |
| Mystery     | 389            | 6.139130435 | 0      | 0    | 0     | -            | -           |
| Romance     | 885            | 6.2         | 0      | 0    | 0     | -            | -           |
| Sci-Fi      | 501            | 5.7375      | 6.5    | 6.7  | 7.9   | 1.125115866  | 1.265885711 |
| Thriller    | 1127           | 6.266666667 | 0      | 0    | 0     | -            | -           |
| Western     | 60             | 5.433333333 | 0      | 0    | 0     | -            | -           |

Descriptive statistics of IMDB scores for each genre

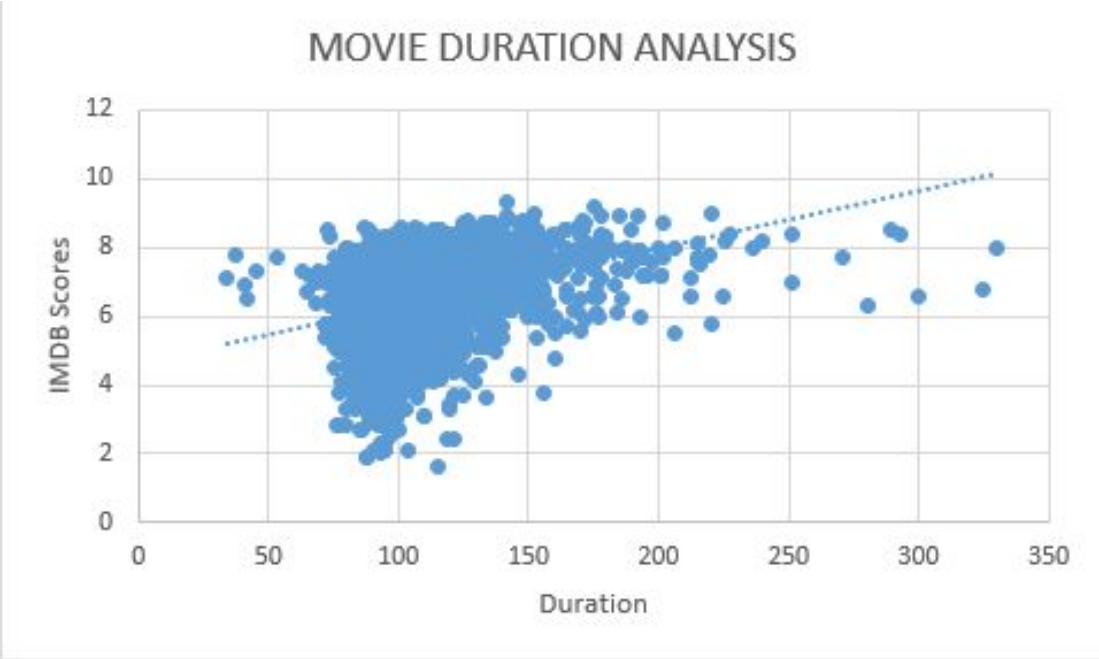
**B. MOVIE DURATION ANALYSIS:**

- The task was to analyze the distribution of movie durations and identify the relationship between movie duration and IMDB score:
  - i. Calculated descriptive statistics such as mean, median, and standard deviation for movie durations.
  - ii. Used Excel's functions like **AVERAGE**, **MEDIAN**, and **STDEV**.

**=AVERAGE(C:C)** calculates the mean duration, providing the central tendency. The **=MEDIAN(C:C)** computes the median duration, representing the middle value. Lastly, the **=STDEV(C:C)** calculates the standard deviation, indicating the dispersion or variability of movie durations in the dataset. Together, they offer insights into the central tendency and spread of movie duration data.

|            |             |
|------------|-------------|
| MEAN       | 107.201074  |
| MEDIAN     | 103         |
| STANDARD D | 25.19744081 |

- iii. Created a scatter plot to visualize the relationship between movie duration and IMDB score. Added a trend line to assess the direction and strength of the relationship.



### C. LANGUAGE ANALYSIS:

- The task was to determine the most common languages used in movies and analyze their impact on the IMDB score using descriptive statistics.

**i. Used Excel's COUNTIF function to count the number of movies for each language.**

The formula **=COUNTIF(J2:J5440, "Aboriginal\*")** counts the number of cells in the range J2 to J5440 that start with the term "Aboriginal." It employs the COUNTIF function in Excel, where the first argument specifies the range to evaluate, and the second argument sets the criteria for counting. The asterisk (\*) functions as a wildcard character, ensuring the count includes cells starting with "Aboriginal" followed by any additional characters.

**ii. Calculated the mean, median, and standard deviation of the IMDB scores for each language.**

The formulas collectively analyze the data in the range J2 to J5440 based on specified criteria.

**'=AVERAGEIF(J2:J5440,AR3,N2:N5440)'** calculates the average of values in N2 TO N5440 where corresponding cells in J2 to J5440 match the criteria in AR3. **'=MEDIAN(J2:J5440,AR3,N2:N5440)'** computes the median using the same criteria, and **'=STDEV.S(J2:J5440,AR3,N2:N5440)'** calculates the standard deviation for the specified subset of data meeting the given condition.

**iii. Combined all the count and descriptive statistics into a table as seen in the next slide**



| LANGUAGES  | COUNT | MEAN        | MEDIAN | S.D         |
|------------|-------|-------------|--------|-------------|
| Aboriginal | 2     | 6.95        | 6.6    | 1.125115866 |
| Arabic     | 5     | 7.38        | 6.6    | 1.125115866 |
| Aramaic    | 1     | 7.1         | 6.6    | 1.125115866 |
| Bosnian    | 1     | 4.3         | 6.6    | 1.125115866 |
| Cantonese  | 11    | 6.954545455 | 6.6    | 1.125115866 |
| Czech      | 1     | 7.4         | 6.6    | 1.125115866 |
| Danish     | 5     | 7.5         | 6.6    | 1.125115866 |
| Dari       | 2     | 7.5         | 6.6    | 1.125115866 |
| Dutch      | 4     | 7.425       | 6.6    | 1.125115866 |
| Dzongkha   | 1     | 7.5         | 6.6    | 1.125115866 |
| English    | 4704  | 6.398107591 | 6.6    | 1.125115866 |
| Filipino   | 1     | 6.7         | 6.6    | 1.125115866 |
| French     | 73    | 7.038356164 | 6.6    | 1.125115866 |
| German     | 19    | 7.342105263 | 6.6    | 1.125115866 |
| Hebrew     | 5     | 7.58        | 6.6    | 1.125115866 |
| Hindi      | 28    | 6.632142857 | 6.6    | 1.125115866 |
| Hungarian  | 1     | 7.1         | 6.6    | 1.125115866 |
| Icelandic  | 2     | 7.55        | 6.6    | 1.125115866 |
| Indonesian | 2     | 7.9         | 6.6    | 1.125115866 |
| Italian    | 11    | 7.227272727 | 6.6    | 1.125115866 |
| Japanese   | 18    | 7.394444444 | 6.6    | 1.125115866 |
| Kazakh     | 1     | 6           | 6.6    | 1.125115866 |
| Korean     | 8     | 7.3875      | 6.6    | 1.125115866 |

|            |    |             |     |             |
|------------|----|-------------|-----|-------------|
| Mandarin   | 26 | 6.788461538 | 6.6 | 1.125115866 |
| Maya       | 1  | 7.8         | 6.6 | 1.125115866 |
| Mongolian  | 1  | 7.3         | 6.6 | 1.125115866 |
| None       | 2  | 7.95        | 6.6 | 1.125115866 |
| Norwegian  | 4  | 7.15        | 6.6 | 1.125115866 |
| Persian    | 4  | 7.575       | 6.6 | 1.125115866 |
| Portuguese | 8  | 7.4875      | 6.6 | 1.125115866 |
| Romanian   | 2  | 7.2         | 6.6 | 1.125115866 |
| Russian    | 11 | 6.363636364 | 6.6 | 1.125115866 |
| Spanish    | 40 | 6.9375      | 6.6 | 1.125115866 |
| Swedish    | 5  | 7.44        | 6.6 | 1.125115866 |
| Telugu     | 1  | 8.4         | 6.6 | 1.125115866 |
| Thai       | 3  | 6.633333333 | 6.6 | 1.125115866 |
| Vietnamese | 1  | 7.4         | 6.6 | 1.125115866 |
| Zulu       | 2  | 7.1         | 6.6 | 1.125115866 |

The most common language used in movies is **English** with count **4704**

## D. DIRECTOR ANALYSIS:

- The task was to identify the top directors based on their average IMDB score and analyze their contribution to the success of movies using percentile calculations.
  - Firstly, identified top 10 directors using a pivot table and applying sort filter.
  - Then calculated average IMDB scores as follows.
  - Then used Excel's PERCENTILE function to identify the directors with the highest scores.

The formula **=LARGE(AY10:AY19,1)** identifies the largest value in the range AY10 to AY19, effectively returning the maximum value. **=PERCENTRANK(AY10:AY19,8.6)** calculates the percentile rank of the value 8.6 within the data range AY10 to AY19, indicating its relative position in the dataset. Finally, **=PERCENTILE(AY10:AY19,AY22)** computes the percentile value corresponding to the percentage specified in cell AY22 within the range AY10 to AY19

| top 10 directors       | avg imbd |
|------------------------|----------|
| Charles Chaplin        | 8.6      |
| Tony kaye              | 8.6      |
| Alfred Hitchcock       | 8.5      |
| Darmien Chazelle       | 8.5      |
| Majid Majidi           | 8.5      |
| Ron Fricke             | 8.5      |
| Sergio Leone           | 8.43     |
| Christopher Nolan      | 8.43     |
| Asghar Farhadi         | 8.4      |
| Marius A. Markevicious | 8.4      |

|              |        |
|--------------|--------|
| LARGE        | 8.6    |
| PERCENT RANK | 0.888  |
| PERCENTILE   | 8.5992 |



## E. BUDGET ANALYSIS:

- The task was to analyze the correlation between movie budgets and gross earnings, and identify the movies with the highest profit margin.
- i. **Calculated the correlation coefficient between movie budgets and gross earnings using Excel's CORREL function.**
- ii. **Calculated the profit margin (gross earnings - budget) for each movie and identify the movies with the highest profit margin using Excel's MAX function.**

The formula **=CORREL(D:D, L:L)** calculates the correlation coefficient between two columns of data, specifically between the "Gross" earnings in column D and the "Budget" values in column L. The correlation coefficient provides a measure of the strength and direction of the linear relationship between the two variables; in this case, it helps analyze the correlation between movie budgets and gross earnings.

The formula **=MAX(BD:BD)** identifies the maximum value in the "Profit" column (column BD). It returns the highest profit recorded among all the entries in the specified column, helping to identify the movie with the greatest profit margin in the dataset.

0.102179454 CORRELATION  
523505847 HIGHEST PROFIT MARGIN

After calculating the highest profit margin

**=LARGE(BD:BD, ROW(BA3))**

**=INDEX(BA:BA, MATCH(BE3, BD:BD, 0))**

These formulas will give the top 10 movies with the highest profit margin along with their corresponding movie names.

#### TOP 10 MOVIES WITH HIGHEST PROFIT MARGINS

Captain America: Civil War

Dirty Grandpa

Pirates of the Caribbean: On Stranger Tides

Pirates of the Caribbean: On Stranger Tides

The Croods

Up

The Hangover Part II

Holy Man

The Twilight Saga: Breaking Dawn - Part 2

Wyatt Earp

### 3. Tech-Stack Used:

- For this project I have used Excel 2016 which is a spreadsheet application in the Microsoft Office 2016 suite. It is used for tasks like data entry, analysis, and visualization. Excel 2016 provides tools for creating tables, charts, and performing calculations. It's a widely used program for managing and analyzing data.

[https://docs.google.com/spreadsheets/d/1j\\_AeaEkzduXdSvZ08d4ocHRArNw8JJXR/edit?usp=sharing&oid=112612761911126686527&rtpof=true&sd=true](https://docs.google.com/spreadsheets/d/1j_AeaEkzduXdSvZ08d4ocHRArNw8JJXR/edit?usp=sharing&oid=112612761911126686527&rtpof=true&sd=true)

**Above is the hyperlink of the excel file that I have worked with.**

### 4. Insights:

The insights and knowledge I have gained while working on this project are as follows:

- Excel formulas automate calculations, saving time and reducing errors compared to manual calculations.
- Excel formulas enable data analysis, summarization, and extraction of insights from large datasets.
- Excel charts provide a visual representation of data, making it easier to understand and interpret complex information.
- Excel charts help in identifying trends, outliers, and correlations within data, which may not be immediately apparent in raw numbers.

### 5. Result:

#### A. Movie Genre Analysis:

- Identified the most common genres in the dataset using COUNTIF to quantify genre occurrences.
- Conducted in-depth genre-specific analyses by calculating descriptive statistics (mean, median, mode, range, variance, standard deviation) for IMDB scores.
- Provided insights into how different genres impact movie ratings through statistical comparisons.

#### B. Movie Duration Analysis:

- Analyzed the distribution of movie durations using descriptive statistics (mean, median, standard deviation).
- Explored the relationship between movie duration and IMDB score using a scatter plot.
- Added a trendline to visually assess the direction and strength of the relationship, offering insights into how movie duration influences ratings.

#### C. Language Analysis:

- Identified the most common languages in movies through COUNTIF analysis.
- Analyzed the impact of language on IMDB scores by calculating mean, median, and standard deviation for each language.
- Provided a nuanced understanding of how language diversity contributes to variations in movie ratings.

#### D. Director Analysis:

- Identified top directors based on average IMDB scores, showcasing their influence on movie success.
- Utilized percentile calculations to compare director scores to the overall distribution, highlighting exceptional directorial contributions.
- Provided insights into how specific directors significantly impact the ratings landscape.

#### E. Budget Analysis:

- Analyzed the correlation between movie budgets and gross earnings using the CORREL function.
- Calculated profit margins for each movie (gross earnings - budget) to identify financial success.
- Leveraged the MAX function to pinpoint movies with the highest profit margins, offering insights into budgetary impact on financial success.

Overall, the project achieved a holistic understanding of various factors influencing movie ratings, encompassing genres, duration, language, directors, and budget. The progression of tasks ensured a comprehensive exploration, unveiling meaningful insights into the dataset. In conclusion, this project provided me with the practical exposure in using Formulas, Tables and Charts present in Excel.